

MIC regression

Lucy + Thel + Rhys

2025-11-06

```
library(tidyverse)
library(cowplot)
library(readxl)
library(viridisLite)

# install_github("lu-harr/iddoPal")
library(iddoPal)
# install_github("lu-harr/looseVis")
library(looseVis)

dat <- read_excel("GAS_SR_Raw_50_90_NOS_formatted_20251106.xlsx")
```

Here are some quick summary stats:

```
# number of isolates
sum(dat$count)
```

```
## [1] 39737
```

```
# number of isolates per country
dat %>%
  group_by(country) %>%
  summarise(n = sum(count)) %>%
  arrange(desc(n))
```

```
## # A tibble: 47 x 2
##   country      n
##   <chr>    <dbl>
## 1 Japan      5984
## 2 Spain      5449
## 3 Serbia     5282
## 4 USA        3968
## 5 Brazil     2710
## 6 Italy      2351
## 7 Worldwide  1485
## 8 Canada     1271
## 9 Austria    1084
## 10 Germany    836
## # i 37 more rows
```

```
# number of isolates per who region
dat %>%
  group_by(who) %>%
  summarise(n = sum(count)) %>%
  arrange(desc(n))
```

```
## # A tibble: 7 x 2
##   who     n
##   <dbl> <dbl>
## 1     4 20814
## 2     2  9024
## 3     6  7040
## 4    NA  2174
## 5     3   331
## 6     1   270
## 7     5    84

# combined mic
mic <- dat %>%
  mutate(nos_90_comb = ifelse(is.na(micnos_1), mic90_1, micnos_1))

which(is.na(mic$nos_90_comb))

## integer(0)

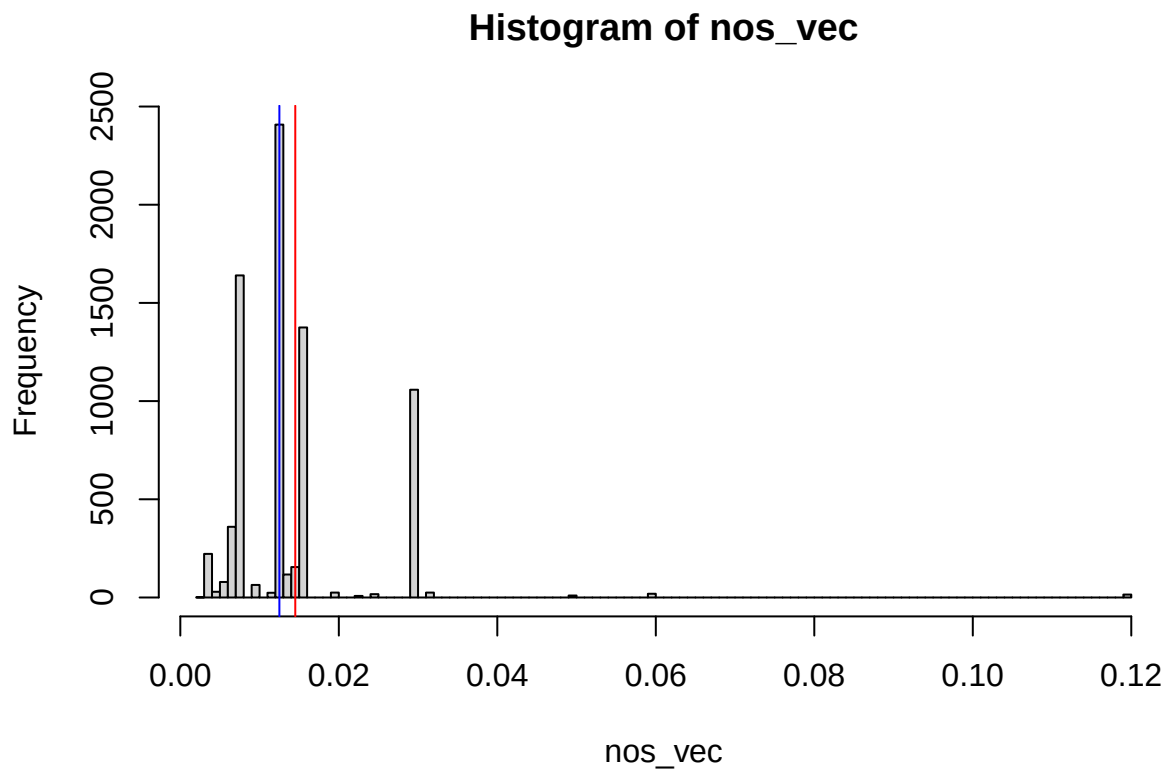
nos_vec <- rep(mic$micnos_1, mic$count)
median(nos_vec, na.rm=TRUE)

## [1] 0.0125

mean(nos_vec, na.rm=TRUE)

## [1] 0.0145049

hist(nos_vec, breaks=100)
abline(v = median(nos_vec, na.rm=TRUE), col="blue")
abline(v = mean(nos_vec, na.rm=TRUE), col="red")
```



```
# represent entries with multiple isolates multiple times:
nos_90_comb_vec <- rep(mic$nos_90_comb, mic$count)
length(nos_90_comb_vec)
```

```
## [1] 39737
```

```
median(nos_90_comb_vec)
```

```
## [1] 0.016
```

```
quantile(nos_90_comb_vec, probs = c(0.1, 0.5, 0.9))
```

```
## 10% 50% 90%
```

```
## 0.008 0.016 0.031
```

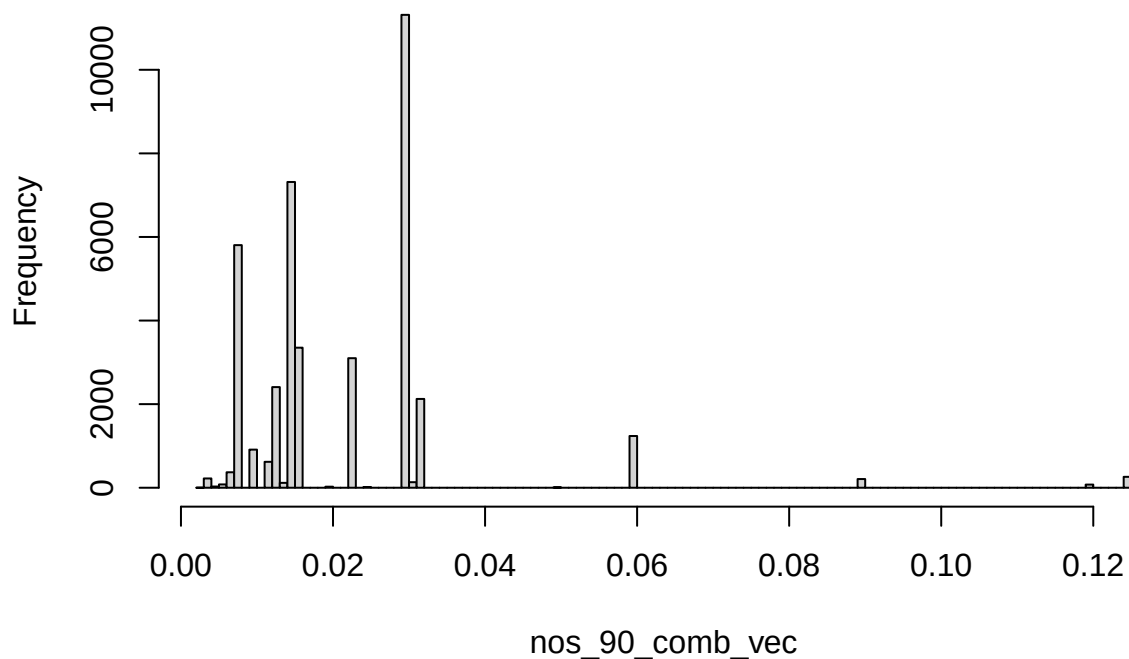
```
summary(nos_90_comb_vec)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
```

```
## 0.00200 0.01250 0.01600 0.02217 0.03000 0.12500
```

```
hist(nos_90_comb_vec, breaks = 100)
```

Histogram of nos_90_comb_vec



And now onto the regression ...

```
who_df <- matrix(c(c("1", "Africa"),
                    c("2", "Americas"),
                    c("3", "Eastern Mediterranean"),
                    c("4", "Europe"),
                    c("5", "Southeast Asia"),
                    c("6", "Western Pacific")),
                 ncol = 2, byrow = TRUE) %>%
```

```

data.frame() %>%
  setNames(c("nume", "name"))

who_lst = who_df$name
names(who_lst) = who_df$nume

# and join to data:
dat <- dat %>%
  mutate(who = as.factor(who)) %>%
  left_join(who_df, by = join_by(who == nume)) %>%
  mutate(who_name = ifelse(is.na(name), "Not assigned", name)) %>%
  # need to put both MIC_nos and MIC_90 in one column
  mutate(mic_mix = case_when(!is.na(micnos_1) ~ micnos_1,
                             !is.na(mic90_1) ~ mic90_1,
                             TRUE ~ NA),
         weight_mic = 1 / mic_mix)

```

Populate names of WHO regions:

```

mod_unweighted = lm(mic_mix ~ isolate_yr, dat)
summary(mod_unweighted)

```

Variance-weighted least squares

```

##
## Call:
## lm(formula = mic_mix ~ isolate_yr, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.019313 -0.009108 -0.005491  0.000399  0.104851
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.165e-01  1.974e-01  -0.590   0.555
## isolate_yr   6.850e-05  9.872e-05   0.694   0.488
##
## Residual standard error: 0.02038 on 275 degrees of freedom
## Multiple R-squared:  0.001748, Adjusted R-squared: -0.001882
## F-statistic: 0.4815 on 1 and 275 DF, p-value: 0.4883

# setting weight to mean absolute residuals
wts <- data.frame(isolate_yr = dat$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat <- dat %>%
  left_join(wts)

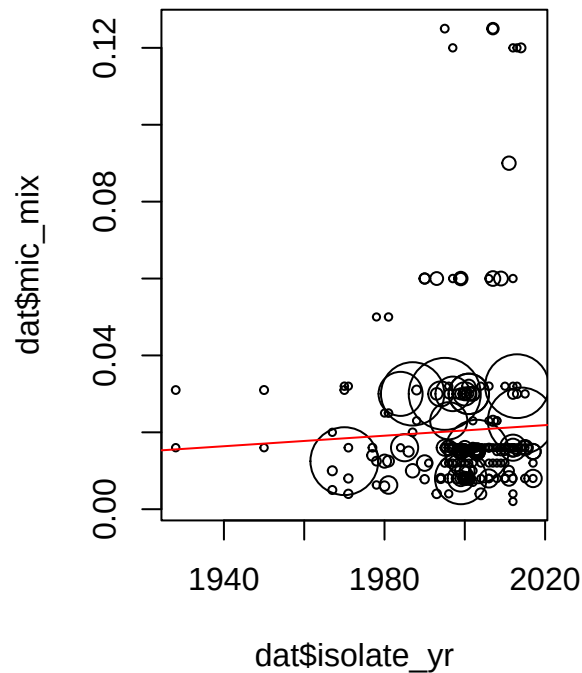
## Joining with `by = join_by(isolate_yr)`

# a quick summary:
par(mfrow = c(1,2))
pred <- predict(mod_unweighted, dat)

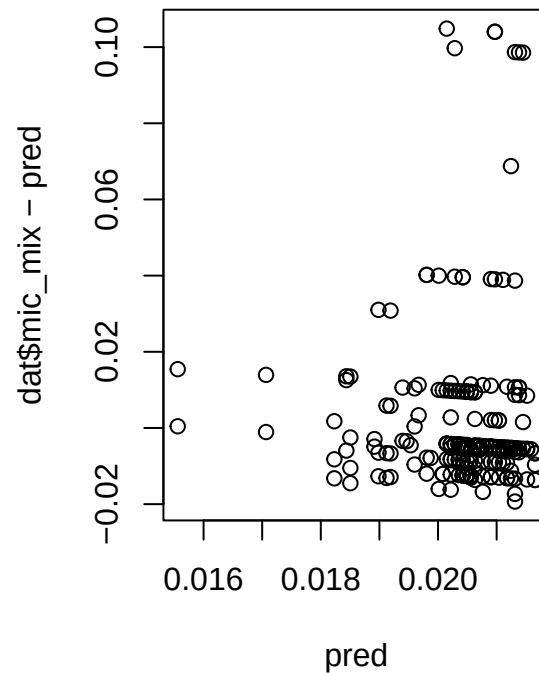
```

```
plot(dat$isolate_yr, dat$mic_mix, cex = scale_cex(dat$count, max_cex = 5), main = "(unweighted) ordinary least squares",
abline(mod_unweighted, col="red")
plot(pred, dat$mic_mix - pred, main = "Residuals over fitted")
```

(unweighted) ordinary least squares



Residuals over fitted

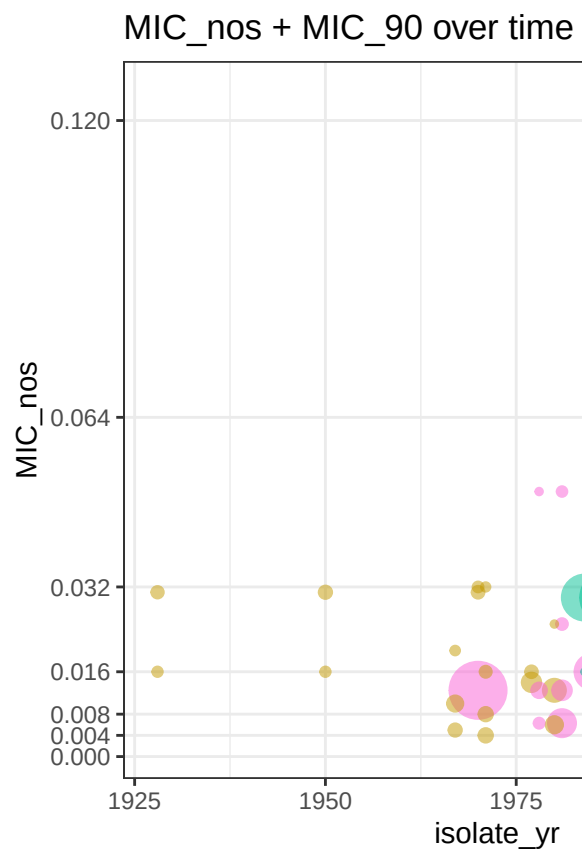


```
# now apply weights to mic_mix
```

```
yticks <- c(0, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  # put smaller points in front:
  arrange(desc(count)),
  aes(x = isolate_yr, y = mic_mix)) +
  geom_point(aes(col = who_name,
    size = count), alpha = 0.5) +
  # pop a line on there
  # geom_smooth(method = "lm", se = FALSE, col = "darkgrey") +
  scale_color_discrete(name = "WHO Region") +
  scale_size_continuous(range = c(1,10), name = "Number of isolates") +
  # set specific ticks on y axis: (could transform this too if we think we need to ...)
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC_nos") +
  labs(title = "MIC_nos + MIC_90 over time (points not grouped)") +
  theme_bw()
```

```
p
```



Here's the bubble plot with both the MIC_nos and the MIC_90s:

```
gap <- 1

# cobble something together ...
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
        iddo_palettes$iddo[1],
        "orange") %>%
  unname()

# model weighted with weights from above
mod_weighted <- lm(mic_mix ~ isolate_yr, dat, weights = wt)
summary(mod_weighted)
```

Now here's a bubble plot with regression:

```
##
## Call:
## lm(formula = mic_mix ~ isolate_yr, data = dat, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0022383 -0.0011114 -0.0006127 -0.0000777  0.0157608
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2025994  0.2272423  -0.892   0.373
```

```
## isolate_yr    0.0001125  0.0001137   0.990    0.323
##
## Residual standard error: 0.00219 on 275 degrees of freedom
## Multiple R-squared:  0.00355,    Adjusted R-squared:  -7.306e-05
## F-statistic: 0.9798 on 1 and 275 DF,  p-value: 0.3231

confint(mod_weighted)

##                2.5 %          97.5 %
## (Intercept) -0.6499550168 0.2447561606
## isolate_yr  -0.0001112752 0.0003363511

tmp <- predict(mod_weighted, data.frame(isolate_yr = 1928:2017), se.fit = TRUE)

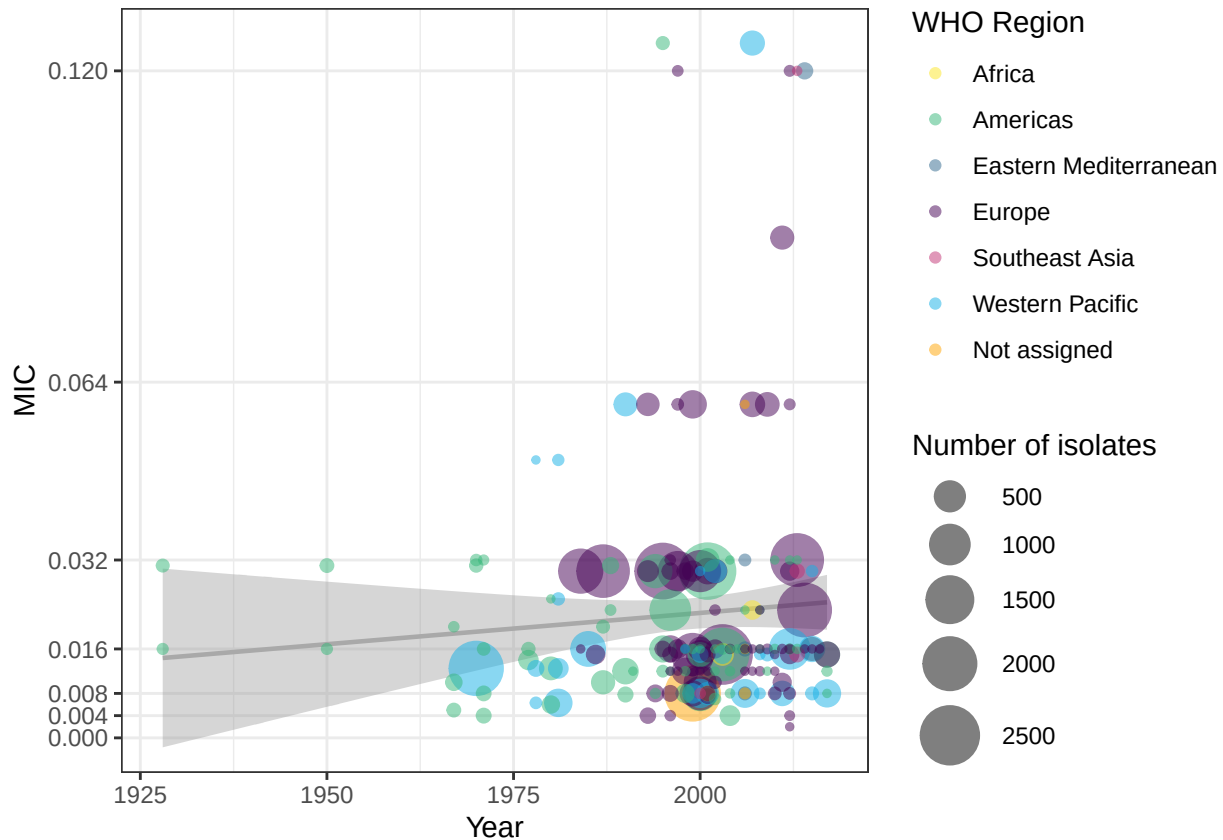
p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  group_by(isolate_yr, mic_mix, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%

  # put smaller points in front:
  arrange(desc(count)),
  aes(x = isolate_yr, y = mic_mix)) +
coord_cartesian(ylim = c(0, 0.125)) +
# pop a line on there
geom_smooth(data = dat %>% filter(!is.na(mic_mix)),
  #mapping = aes(weight = count),
  #mapping = aes(weight = weight_mic),
  mapping = aes(weight = wt),
  method = "lm", se = TRUE, col = "darkgrey", linewidth = 0.8) +
#geom_smooth() +
geom_point(aes(col = who_name,
  size = count), alpha = 0.5) +
scale_color_discrete(name = "WHO Region", type = pal) +
# scale_size_continuous(range = c(1,6), name = "Number of isolates",
#                               trans="log10") +
scale_size_continuous(range = c(1,10), name = "Number of isolates") +
# set specific ticks on y axis: (could transform this too if we think we need to ...)
scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC#",
  #limits = c(0, 0.125)
) +
scale_x_continuous(limits = c(1928 - gap, 2017 + gap)) +
#labs(title = "MIC_nos + MIC_90 over time (points grouped, variance-weighted LS)") +
labs(x = "Year", ylab = "MIC") +
theme_bw()

## `summarise()` has grouped output by 'isolate_yr', 'mic_mix'. You can override
## using the `.groups` argument.
```

p

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
ggsave("figs/mic_mixed_bubbles.png",  
        height = 6, width = 9, scale = 0.8)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
# would love to transform the y axis but if i can't extract the ribbon from  
# geom_smooth then I can't tell it not to freak out at zero ..
```

```
# sad - doesn't look like coord_cartesian will work with axis breaks  
# https://github.com/YuLab-SMU/ggbreak/issues/59  
library(ggbreak)
```

Here's the same plot with some breaks in the x axis:

```
## ggbreak v0.1.6 Learn more at https://yulab-smu.top/
```

```
## If you use ggbreak in published research, please cite the following
```

```
## paper:
```

```
##
```

```
## S Xu, M Chen, T Feng, L Zhan, L Zhou, G Yu. Use ggbreak to effectively
```

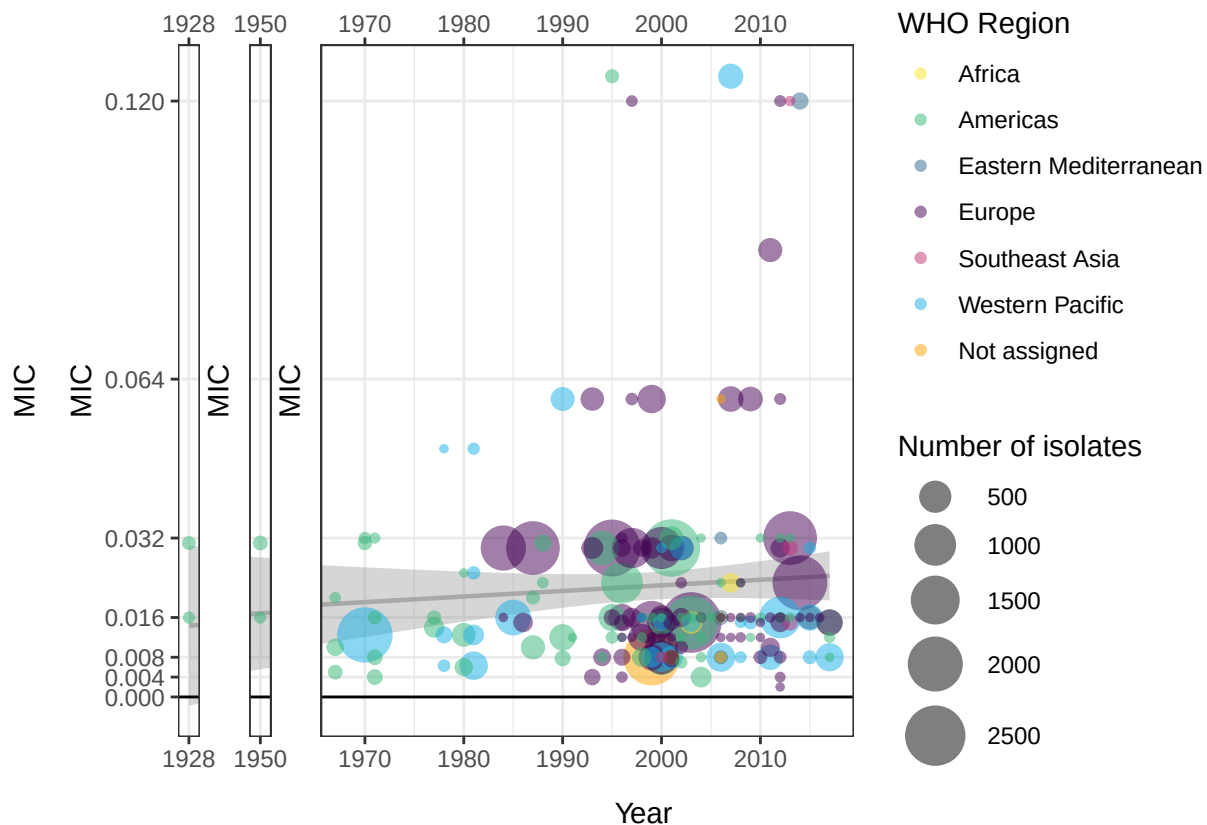
```
## utilize plotting space to deal with large datasets and outliers.
```

```
## Frontiers in Genetics. 2021, 12:774846. doi: 10.3389/fgene.2021.774846
```

```
p + scale_x_break(c(1928 + gap, 1950 - gap)) +
  coord_cartesian(ylim = c(0, 0.125)) +
  scale_x_break(c(1950 + gap, 1968)) +
  scale_x_continuous(breaks = c(1928, 1950, seq(1960, 2021, 10)),
                    limits = c(1928 - gap, 2016 + gap)) +
  geom_hline(aes(yintercept = 0)) +
  xlab("Year") + ylab("MIC")
```

```
## Coordinate system already present. Adding new coordinate system, which will
## replace the existing one.
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



```
ggsave("figs/mic_mixed_bubbles_with_breaks.png",
       height = 6, width = 9, scale = 0.8)
# make colours viridis plus an orange maybe?
```

Now here's a plot with the y-axis logged but *not* the regression: (so regression's still on a linear scale)

```

Y_LOWER <- 0.0018
X_LOWER <- 1925
mod_weighted <- lm(mic_mix ~ isolate_yr, dat, weights = wt)
confint(mod_weighted)

##                2.5 %          97.5 %
## (Intercept) -0.6499550168 0.2447561606
## isolate_yr  -0.0001112752 0.0003363511

ci <- predict(lm(mic_mix~isolate_yr, dat, weights = wt),
              newdata = data.frame(isolate_yr = seq(X_LOWER,
                                                    max(dat$isolate_yr) + 2)),
              interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))

# okay: need to transform y-axis and truncate
gap <- 1

# cobble something together ...
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
        iddo_palettes$iddo[1],
        "orange") %>%
  unname()

yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

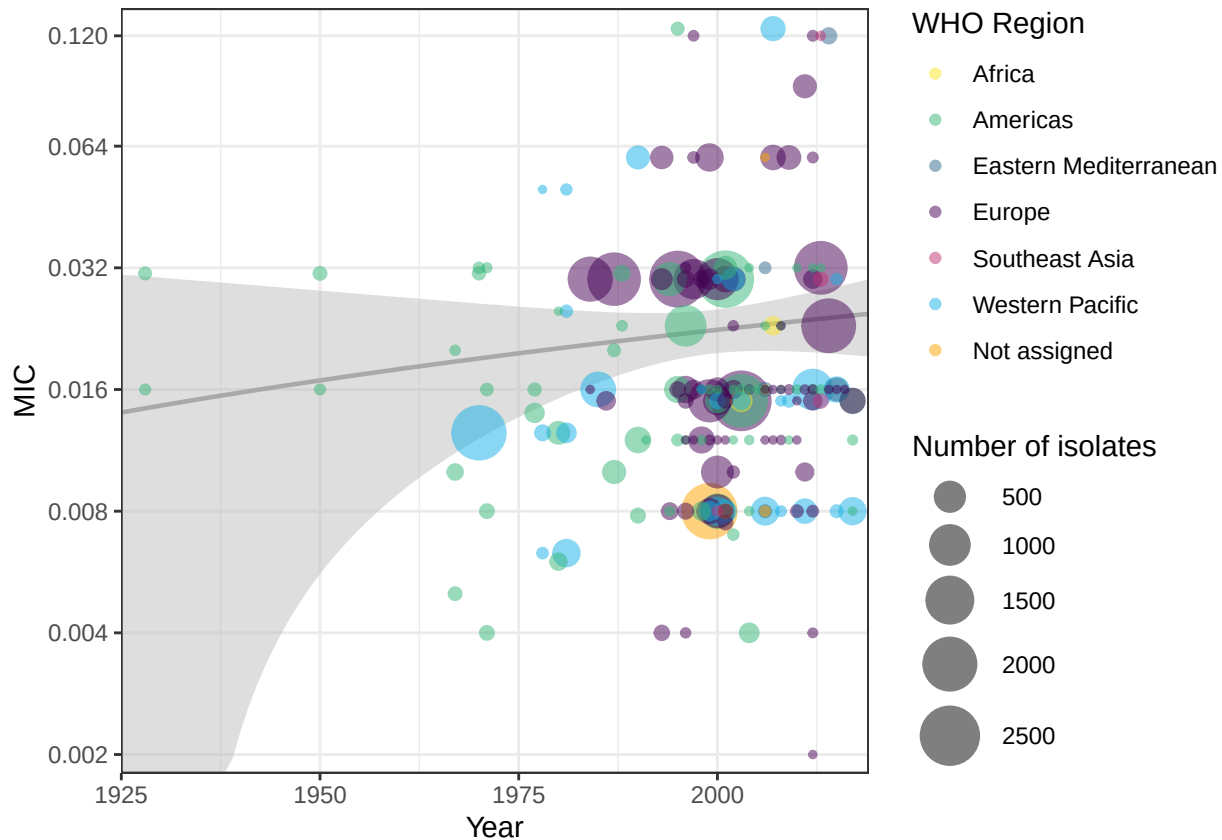
library(ggbreak)
p <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  group_by(isolate_yr, mic_mix, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%
  arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = lwr, ymax = upr, x = isolate_yr),
            alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_mix,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates") +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2)) +
  labs(x = "Year", ylab = "MIC") +

```

```
coord_trans(y = "log2", ylim = c(Y_LOWER, 0.14), expand = FALSE) +
theme_bw()
```

```
## `summarise()` has grouped output by 'isolate_yr', 'mic_mix'. You can override
## using the `.groups` argument.
```

p



```
ggsave("figs/mic_mixed_bubbles_logged.png",
height = 6, width = 9, scale = 0.8)
```

```
# apply log to the mics:
dat <- dat %>%
  mutate(mic_logged = log2(mic_mix))

mod_unweighted = lm(mic_logged ~ isolate_yr, dat)
summary(mod_unweighted)
```

Alternatively, do the regression on the log scale and re-exp the labels

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.99224 -0.40633 -0.07964  0.04772  2.98163
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.9308585  9.0673854  -0.764    0.445
## isolate_yr   0.0004758  0.0045351   0.105    0.917
##
## Residual standard error: 0.9363 on 275 degrees of freedom
## Multiple R-squared:  4.003e-05, Adjusted R-squared:  -0.003596
## F-statistic: 0.01101 on 1 and 275 DF,  p-value: 0.9165

# setting weight to mean absolute residuals
wts <- data.frame(isolate_yr = dat$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat <- dat %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr, wt)`

mod_weighted <- lm(mic_logged ~ isolate_yr, dat, weights = wt)
confint(mod_weighted)

##           2.5 %      97.5 %
## (Intercept) -29.453208006  9.89141900
## isolate_yr   -0.007913285  0.01177093

summary(mod_weighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.25675 -0.05960 -0.01067  0.00858  0.45078
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.780895  9.992906  -0.979    0.329
## isolate_yr   0.001929  0.004999   0.386    0.700
##
## Residual standard error: 0.09629 on 275 degrees of freedom
## Multiple R-squared:  0.000541, Adjusted R-squared:  -0.003093
## F-statistic: 0.1488 on 1 and 275 DF,  p-value: 0.6999

# this is slope on log-scale:
coefficients(mod_weighted)[2]

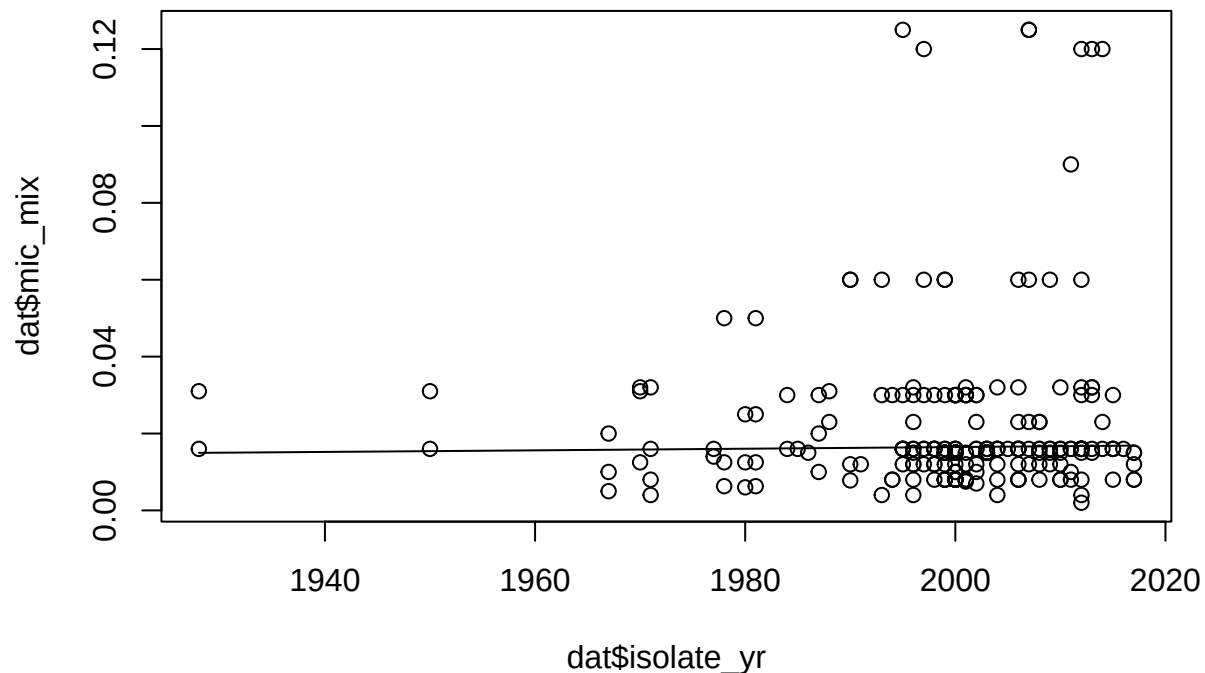
## isolate_yr
## 0.001928824

2*(coefficients(mod_weighted)[2])

## isolate_yr
## 1.001338
```

```
# Here's what that looks like unlogged:
```

```
plot(dat$isolate_yr, dat$mic_mix)
yyy = seq(min(dat$isolate_yr), max(dat$isolate_yr))
lines(yyy, 2**predict(mod_weighted, data.frame(isolate_yr = yyy)))
```



```
Y_LOWER <- log2(0.0018)
X_LOWER <- 1925

ci <- predict(lm(mic_logged ~ isolate_yr, dat, weights = wt),
              newdata = data.frame(isolate_yr = seq(X_LOWER,
                                                    max(dat$isolate_yr) + 2)),
              interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))

gap <- 1

# cobble something together ...
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
        iddo_palettes$iddo[1],
        "orange") %>%
  unname()

yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

p <- ggplot(dat %>%
            filter(!is.na(mic_mix)) %>%
            group_by(isolate_yr, mic_mix, who_name) %>%
            summarise(count = sum(count)) %>%
```

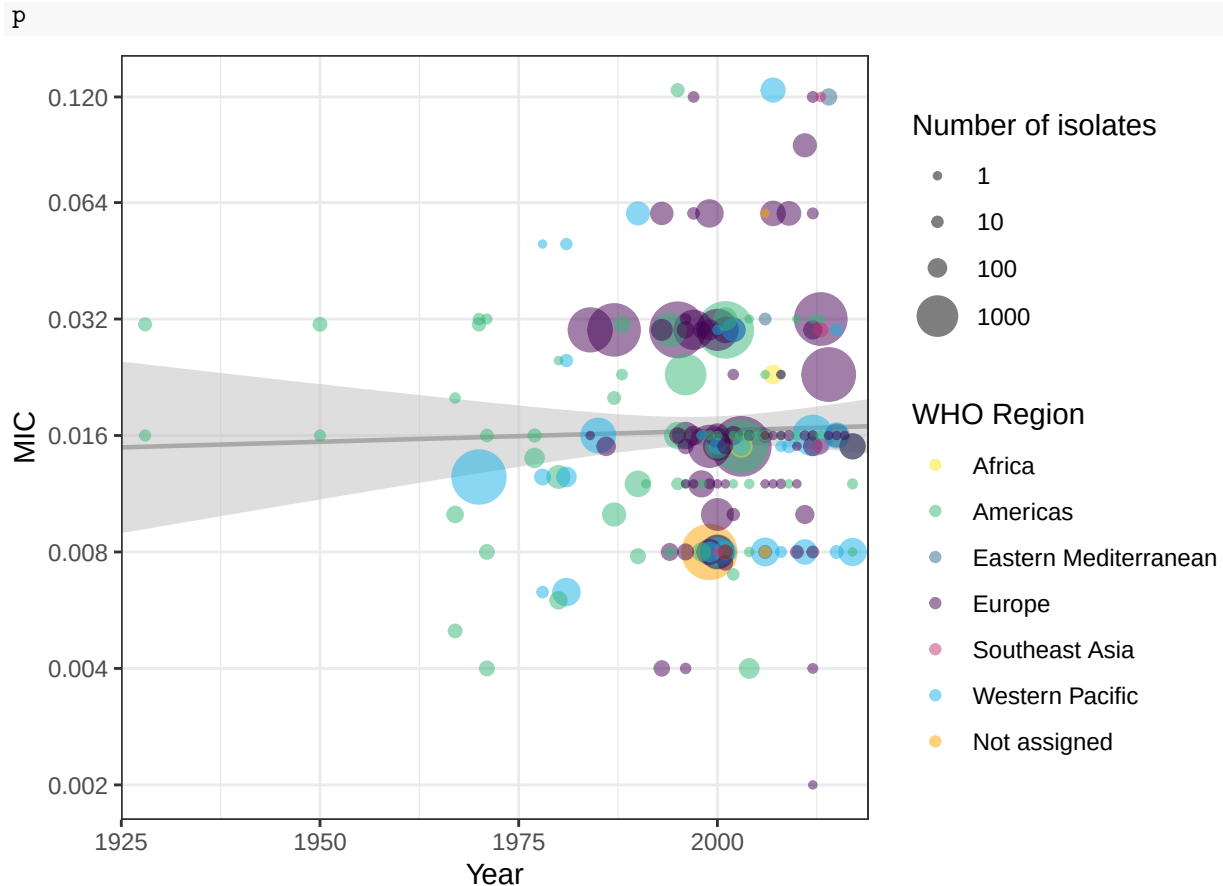
```

mutate(who_name = factor(who_name, levels = c("Africa",
                                              "Americas",
                                              "Eastern Mediterranean",
                                              "Europe",
                                              "Southeast Asia",
                                              "Western Pacific",
                                              "Not assigned")))) %>%

  arrange(desc(count)) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
            alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_mix,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = "pal") +
  scale_size_continuous(range = c(1,10), name = "Number of isolates", breaks = c(1, 10, 100, 1000)) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0)) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2") + #, ylim = c(Y_LOWER, 0.14), expand = FALSE) +
  theme_bw()

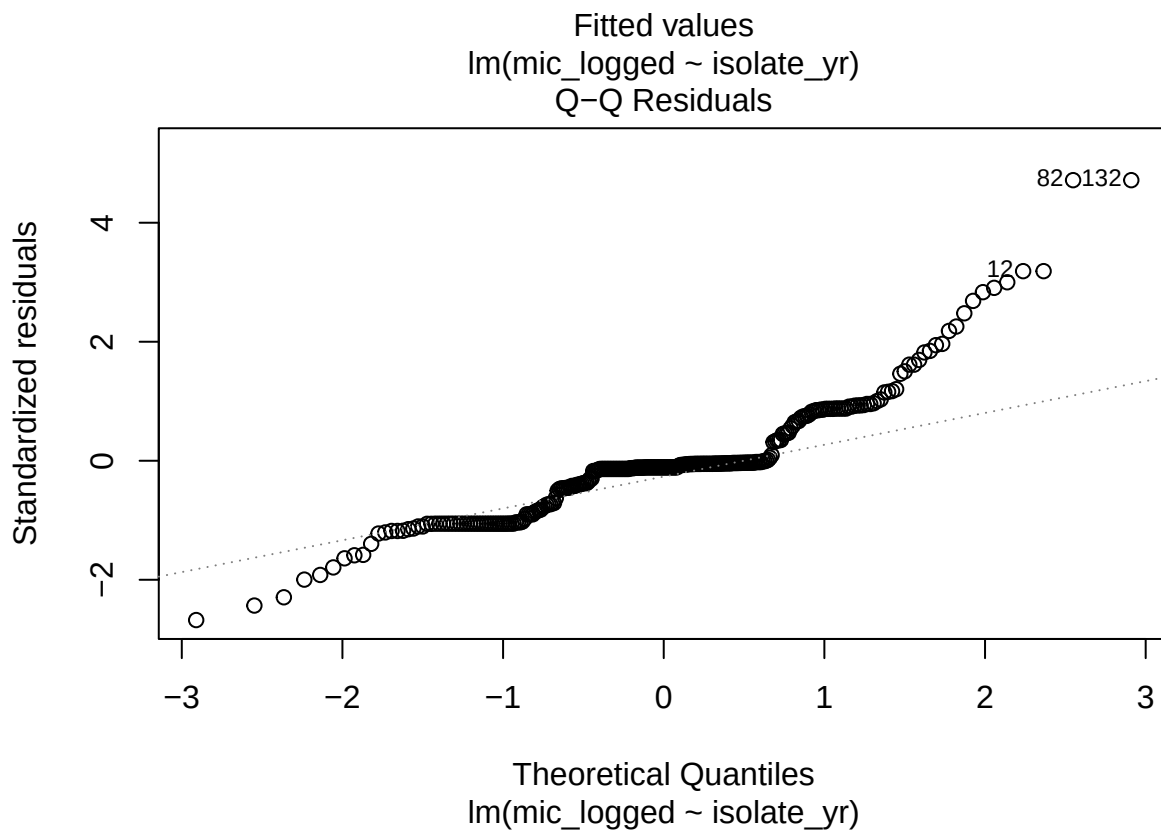
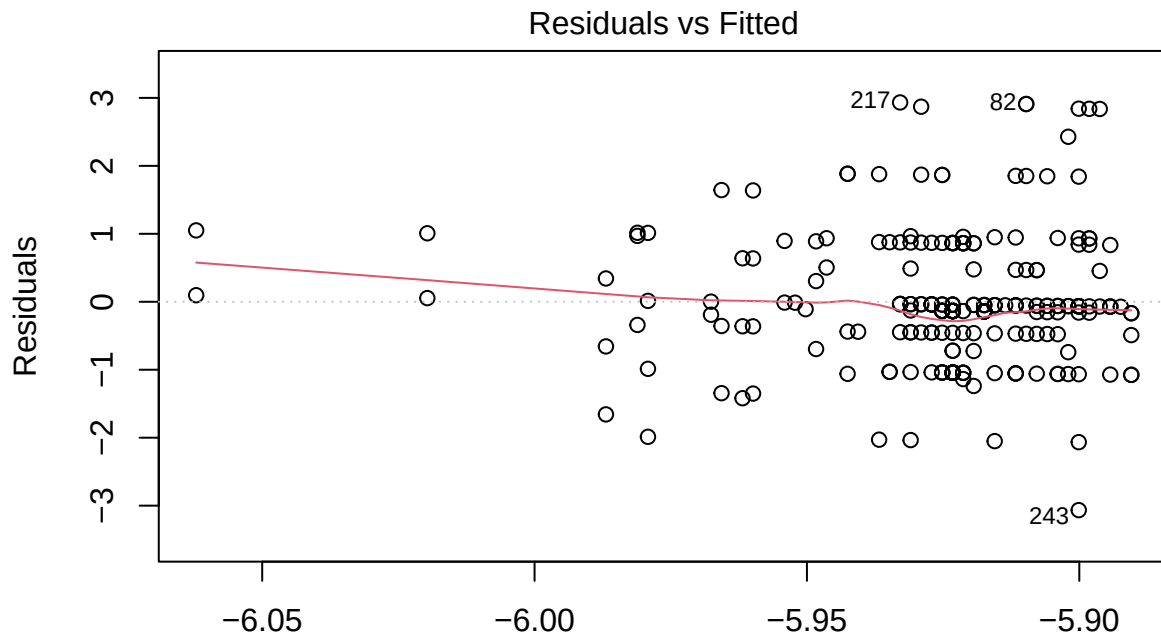
```

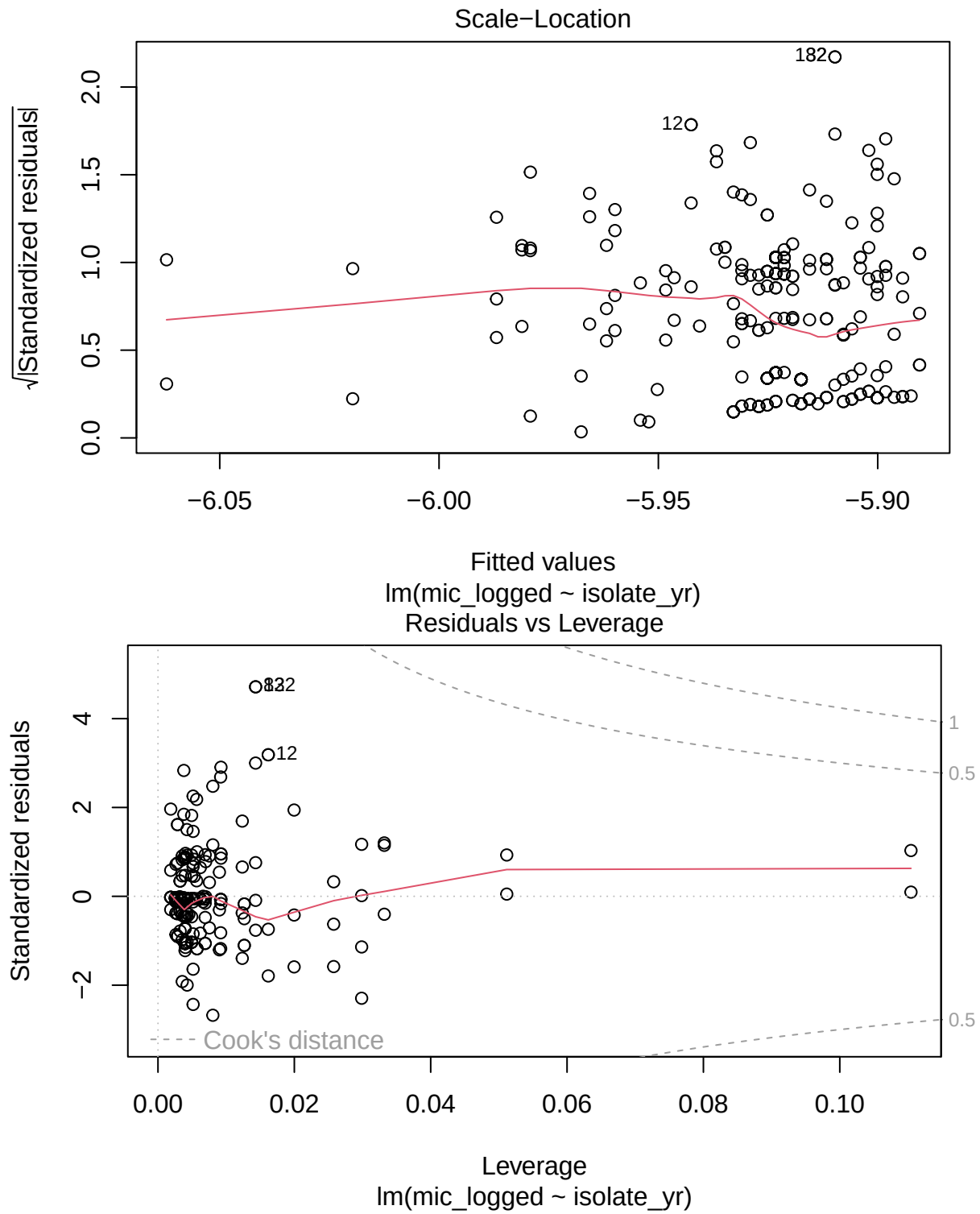
`summarise()` has grouped output by 'isolate_yr', 'mic_mix'. You can override
using the `.groups` argument.



```
ggsave("figs/mic_mixed_bubbles_all_logged.png",
       height = 6, width = 9, scale = 0.8)
```

```
plot(mod_weighted)
```





```
library(MASS)
```

Ordinal regression coming up down here!

```
##  
## Attaching package: 'MASS'
```

```

## The following object is masked from 'package:dplyr':
##
##      select
fable(dat$mic_mix)

## 0.002 0.004 0.005 0.006 0.0063 0.007 0.0075 0.0078 0.008 0.01 0.012 0.0125 0.014 0.015 0.016 0.02 0
##
##      1      5      1      1      2      1      1      1      49      6      19      4      1      57      59      2
# say we bin MIC ..?
bins <- c(0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)
dat$mic_mix_binned <- cut(dat$mic_mix,
                          breaks = c(0, bins, max(dat$mic_mix)),
                          ordered_result = TRUE)
# I've totally forgotten how we got to MIC90s and MICNosses so this ^ may not be
# appropriate

fable(dat$mic_mix_binned)

## (0,0.002] (0.002,0.004] (0.004,0.008] (0.008,0.016] (0.016,0.032] (0.032,0.064] (0.064,0.12] (0.12,0
##
##      1      5      56      146      49      12      5
# Fit the model - no need to log: that would be for visualisation purposes only under ordinal model
# tried using wts from logged model ..... not thinking too hard about that but we should at some po
# it looks like weights need to be integers as we end up assuming binomial glm for ordinal *logistic* r
# haven't thought too hard about this ... may need to be probit link
model <- polr(mic_mix_binned ~ isolate_yr, data = dat, Hess = TRUE)

summary(model)

## Call:
## polr(formula = mic_mix_binned ~ isolate_yr, data = dat, Hess = TRUE)
##
## Coefficients:
##              Value Std. Error t value
## isolate_yr -0.01635  0.0002099  -77.88
##
## Intercepts:
##              Value      Std. Error t value
## (0,0.002] | (0.002,0.004]    -38.1391      0.0059 -6453.4792
## (0.002,0.004] | (0.004,0.008]   -36.5784      0.0318 -1150.3705
## (0.004,0.008] | (0.008,0.016]   -33.9443      0.3947  -85.9931
## (0.008,0.016] | (0.016,0.032]   -31.5681      0.4197  -75.2133
## (0.016,0.032] | (0.032,0.064]   -30.1144      0.4694  -64.1617
## (0.032,0.064] | (0.064,0.12]    -29.1444      0.5405  -53.9167
## (0.064,0.12] | (0.12,0.125]    -28.1691      0.6902  -40.8135
##
## Residual Deviance: 729.6694
## AIC: 745.6694
# note the slope here

new_data = data.frame(isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))
preds <- predict(model,
                  newdata = new_data,

```

```

      type = "probs") %>%
as.data.frame() %>%
mutate(max_probs = max.col(across(starts_with("("))) %>%
mutate(isolate_yr = new_data(isolate_yr)

preds_long <- pivot_longer(preds, starts_with("(")) %>%
  mutate(lower = str_extract(name, "(?<=\\()[^,]+") %>% as.numeric(),
         upper = str_extract(name, "(?<=,)[^]]+") %>% as.numeric(),
         height = lower + (upper - lower)*value) %>%
  filter(lower != 0)

# so now we have colour bars representing class probability between the lower
# and upper extents of the bins ... which don't change much because there isn't
# much change in MIC slope over time
ggplot(data = dat) +
  geom_point(aes(x = isolate_yr, y = mic_mix)) +
  scale_y_continuous(trans = "log2", breaks = bins) +
  geom_ribbon(data = preds_long, aes(ymin = lower, ymax = height, fill = name, x = isolate_yr),
            alpha = 0.5) +
  scale_fill_discrete("MIC bin") +
  # and this black line is the predicted class, which stays the same for the duration of the data
  # (the -1 is because I removed the bottom bin)
  geom_hline(yintercept = bins[preds$max_probs[1] - 1])

```

